points aren’t the same, or checking that the circles used to draw the vertices have no intersection. If the check fails, the embedding is to be rejected.

#4 Cycle through all graph’s edges.

#5 For each edge, cycle through all vertices, to make sure the vertex is not drawn crossing the edge.

#6 If the vertex is not one of the edge’s endpoints, we need to make sure that the edge is not drawn over the vertex (otherwise, it might seem it’s two edges adjacent to this vertex). If this happens for any vertex/edge, we need to reject the embedding.

The implementation of the helper methods used in listings 16.2 and 16.3 was discussed in the last chapter, section 15.4.

Notice that the helper methods checking if a vertex intersects another vertex or an edge are context-dependent, based on how we draw vertices: if they are drawn with circles, we need to make that the circles used to draw every pair of vertices have no intersection, and that no edge is drawn over the vertex’s circle (otherwise, it might seem it’s two edges adjacent to this vertex).

Listing 16.3 Method validateEmbedding

```
function edgeIntersections(graph, embedding)
                              #1
  m ←
|graph.edges|
              #2
  k ← 0
  for i in {0..m-2} do
                  #3
    edge1 ← graph.edges[i]
      for j in {i+1..m-1} do

        edge2 ← graph.edges[j]
```